



SIMATS
ENGINEERING



SIMATS
Saveetha Institute of Medical And Technical Sciences
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

Lab experiment-28

Experiment 28. Forward Chaining

Aim

To implement **Forward Chaining** using **Prolog**.

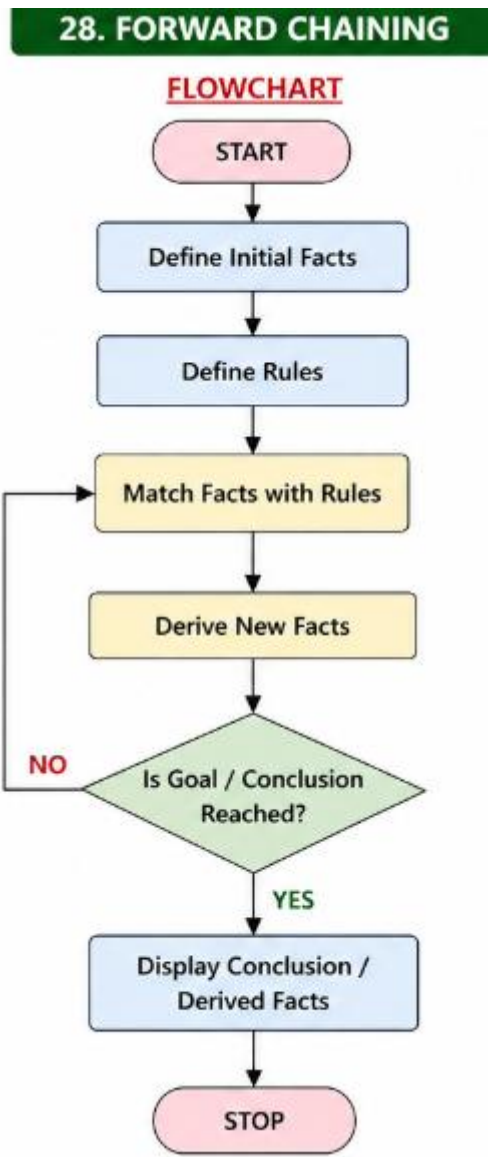
Objectives

- To understand forward chaining.
- To derive new facts from existing facts.
- To perform logical reasoning.

Algorithm

1. Define facts.
2. Define rules.
3. Match facts with rules.
4. Derive new facts.
5. Display the conclusion.

Flowchart



Code

bird(parrot).

has_wings(parrot).

can_fly(X) :-

bird(X),

has_wings(X).

main :-

X = parrot,

can_fly(X),

write('Bird: '), write(X),

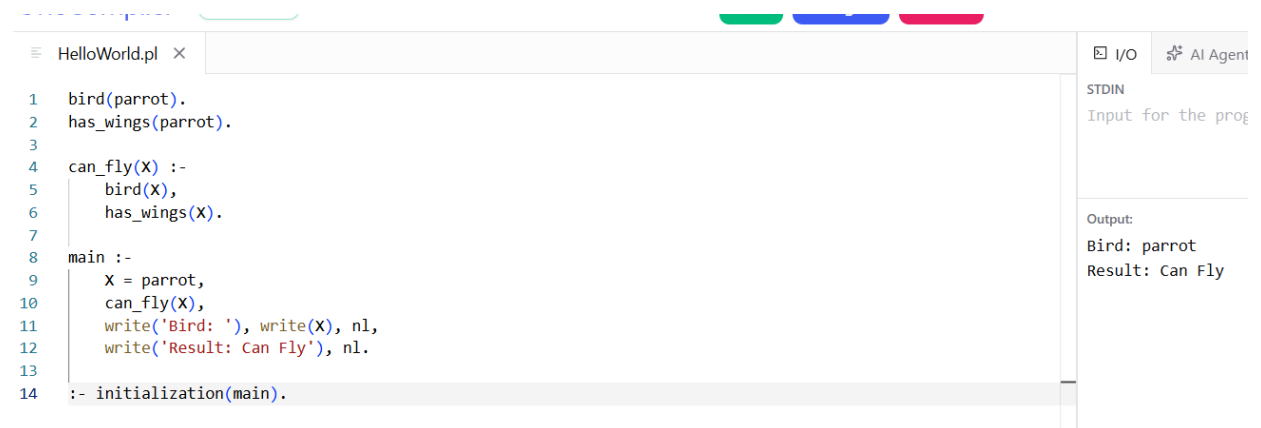
nl,

write('Result: Can Fly'),

nl.

:- initialization(main).

Output



The screenshot shows a Prolog IDE window titled 'HelloWorld.pl'. The code is as follows:

```
1 bird(parrot).
2 has_wings(parrot).
3
4 can_fly(X) :-
5     bird(X),
6     has_wings(X).
7
8 main :-
9     X = parrot,
10    can_fly(X),
11    write('Bird: '), write(X), nl,
12    write('Result: Can Fly'), nl.
13
14 :- initialization(main).
```

On the right side, there is a panel with 'I/O' and 'AI Agent' icons. Under 'STDIN', it says 'Input for the prog'. Under 'Output', it shows the results of the program execution:

```
Output:
Bird: parrot
Result: Can Fly
```

Result

Bird: parrot

Result: Can Fly

Conclusion

Thus, **Forward Chaining** was successfully implemented using Prolog.